

Security-Aware Workload-State Counterfactual Verification and Nodal Net-Value Settlement for Spatially Coupled AI Data Centers

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Abstract—Artificial-intelligence (AI) data centers are flexible grid loads, yet a meter-only baseline cannot establish whether an apparent demand-response (DR) reduction is supported by queued work or by a spatial and temporal redistribution. This paper develops an auditable verifier that carries a committed workload ledger through a feasible service trajectory, a contingency-constrained network-value calculation, and a settlement rule. Its sparse multi-period linear program enforces release, conservation, deadlines, capacity, migration, and cross-midnight completion; a validation-only convex risk fit and an exact pointwise workload envelope separate statistical fitting from the contractual cap. At an event gate, post-gate arrivals are masked before target construction and optimization. Gross forecast credit is then settled only at the pointwise intersection with the event meter, while a causal reserve is reported separately for planning. Across 54 locked days containing 5,188,507 BurstGPT requests and 71,128 positive-energy jobs from the Massachusetts Institute of Technology (MIT) SuperCloud scheduler-Data Center GPU Manager (DCGM) join, the risk-constrained verifier records 1.826322 MWh/day of unsupported gross credit versus 1.849211 for the selected single feasible cap (nRMSE 0.345019 versus 0.345209). The gate-causal rolling service retains 0.995 MWh/day of meter-capped payable response, with zero meter-capped false payment. Exact single-contingency N-1 settlement reduces trace-anchored error from \$165.288 to \$1.018/day over 37 RTS-24 outages, whereas independent end-to-end errors remain \$159.514–\$159.483/day. Job-level residuals are below 10^{-17} MWh. Held-out conversion endpoints, four public alternating-current (AC) networks, all spatial permutations, and a continuous workload-hull interval audit define the reproducibility boundary; the non-co-located traces are a benchmark, not a utility field event.

Index Terms—AI data centers, counterfactual verification, demand response, security-constrained dispatch, nodal settlement.

I. INTRODUCTION

The growth of large-model training and inference is converting data centers from nearly static industrial loads into large, software-defined grid resources. Interactive inference is latency constrained, elastic inference can be delayed briefly, and batch graphics-processing-unit (GPU) work can be deferred for hours or migrated between regions. These capabilities enable demand response (DR), but they also invalidate the usual interpretation of a meter decrease. A reduction at one facility can be followed by delayed consumption or simultaneous rebound at another node. Paying the positive local decrease can therefore reward a redistribution that creates little or negative network value.

Conventional DR measurement and verification estimates the unobserved no-event demand from historical meter data. Arithmetic historical baselines remain common in market practice under published market guidance [1], despite known bias and manipulation incentives [2]. Regression-based facility models quantify baseline uncertainty but do not encode a computing workload ledger [2]. More broadly, DR research establishes the economic role of responsive load, yet a meter-only estimator cannot determine whether queued jobs could actually have produced its predicted trajectory.

Data-center scheduling studies provide the missing physical state. Temporal deferral, dynamic capacity, and geographical load balancing can reduce energy cost and peak demand [3]. Geographical assignment provides the complementary spatial control mechanism [4]. Recent work assesses data-driven flexibility from production traces and system-level security-constrained coordination [5], [6], while aggregator coordination and multi-level dispatch [7], [8] refine delay-sensitive electrical response. Hierarchical optimal-consumption models further connect facility efficiency with response quality [9]. Virtual-link market clearing and space-time remuneration establish how geographical and temporal transfers can enter electricity markets [10], [11]. Receding-horizon operation and uncertainty-aware frequency-regulation coordination further establish the need to retain future arrivals and multi-site state [12]. However, these methods optimize an operator-controlled schedule rather than verify an independently submitted DR counterfactual against an auditable workload ledger. This separation between computing feasibility and market verification motivates the auditable ledger used here. Recent whole-facility H100 measurements likewise require a separate bottom-up conversion from GPU power to facility demand [13]; the present study therefore keeps raw energy calibration, benchmark scaling, and network placement as separate evidence layers.

The second gap is electrical. Geographic workload control is commonly valued with energy prices or gross megawatts. Such accounting ignores the fact that nodal marginal value changes when congestion or a contingency constraint binds. Direct-current (DC) power flow and linear distribution factors provide transparent network sensitivities [14], [15]; N-1 dispatch can impose every credible post-contingency line limit through line-outage distribution factors (LODFs) [16]. No existing data-center DR verification framework jointly certifies workload feasibility, preserves signed spatial response, and settles the resulting security-aware network value. Alternating-current (AC) feasibility is a distinct nonlinear check, not a consequence of the DC certificate. Public workload and facility telemetry can support an observational replay of that ledger, but it does not constitute a causal utility event; a rigorous study must therefore separate measured-execution replay, counterfactual construction, and network-value verification rather than present them as one field intervention.

This paper makes three power-system contributions. 1) We formulate a provenance-bound workload-state demand model in which release, deadline, conservation, cross-midnight completion, capacity, and migration constraints are enforced simultaneously. The resulting job-level witness links each joined job to its immutable scheduler/DCGM record and preserves measured execution energy. 2) We construct a validation-only, continuous-arrival counterfactual verifier with a pointwise credit envelope. The output is an attainable spatial-

temporal load trajectory rather than an unconstrained meter-only prediction, and its statistical performance is evaluated under an explicit false-credit/under-credit tradeoff. 3) We develop signed space–time nodal remuneration and a bilateral net-value contract under preventive N–1 dispatch, and map the feasible workload set into a payment uncertainty interval over held-out workload-to-power conversion. The novelty is the coupling invariant that carries one committed workload state into physical verification, network valuation, and a set-valued settlement rule; the individual LP, dispatch, and allocation primitives are standard and are not claimed as new in isolation. The certificate is conditional on trusted scheduler and telemetry provenance: a cryptographic digest verifies record integrity after commitment but does not authenticate semantically false fields.

The remainder of the paper is organized as follows. Section II defines the spatial workload and network settlement problem. Section III presents the verifier, security-aware value rule, and allocation mechanism. Section IV specifies the data, baselines, and locked protocol. Section V reports the core counterfactual, execution-replay, job-level, security, payment, continuous-horizon, and provenance experiments. Section VI concludes with the deployment boundary and future validation requirements.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. Spatial workload state

Let $\mathcal{S}$, $\mathcal{K}$, $\mathcal{D}$, and $\mathcal{T}$ denote source regions, workload classes, destination data centers, and 15-minute intervals; the symbol h is reserved for a calendar day in statistical panels. The measured ledger supplies arrival energy a_{skt} for source s , class k , and release interval t . The decision $x_{skd\tau} \geq 0$ is the energy served at destination d in interval τ . Its cumulative state is

$$y_{skt} = \sum_{\tau=0}^t \sum_{d \in \mathcal{D}} x_{skd\tau}. \quad (1)$$

With cumulative arrivals $A_{skt} = \sum_{\tau=0}^t a_{sk\tau}$ and deadline D_k , release and deadline feasibility are

$$y_{skt} \leq A_{skt}, \quad (2)$$

$$y_{skt} \geq A_{sk,t-D_k}, \quad t \geq D_k. \quad (3)$$

The state recursion equals service, terminal service equals total arrivals, and $\sum_{s,k} x_{skdt} \leq \bar{P}_d \Delta t$. Facility demand is

$$p_{dt} = P_d^{\text{fix}} + \Delta t^{-1} \sum_{s,k} x_{skdt}. \quad (4)$$

For rolling window end T^+ and final event-day release T^0 , the terminal state is not forced to consume unexpired future work. Instead,

$$y_{skT^+} = A_{sk, \max\{T^0, T^+ - D_k\}}, \quad (5)$$

which completes every job due inside the window and every event-day arrival, while retaining later real arrivals in the release constraints. The window starts $D_{\max} + 96$ intervals before the event day and ends $D_{\max}$ intervals after it; omitted earlier work must therefore expire before the evaluated day. Equations (2) and (3) prevent early execution and late completion, while (4) maps the feasible service state to the facility meter. This model follows workload deferral and geographical assignment foundations [3], [4]; its rolling implementation follows the state-retention principle in [12]. Waiting and migration costs are $w_k(\tau - t)x_{skd\tau}$ and $m_{sd}x_{skd\tau}$, respectively. Because arrivals are fixed, replacing waiting duration by service time adds only a constant.

For provenance, each retained job is represented by the canonical tuple and its digest in

$$\begin{aligned} \ell_j &= (\text{id}_j, t_j^{\text{submit}}, t_j^{\text{start}}, t_j^{\text{end}}, E_j, n_j^{\text{tel}}, q_j, \kappa_j, \sigma_j), \\ \mathcal{L} &= \text{sort}\{\ell_j : j \in \mathcal{J}\}, \\ h_{\mathcal{L}} &= \text{SHA256}(\text{serialize}(\mathcal{L})). \end{aligned} \quad (6)$$

where E_j is the positive aggregated Data Center GPU Manager (DCGM) energy and n_j^{tel} is its raw telemetry-record count. The digest uses the Secure Hash Standard [17]; the canonical row order, one-to-one join, release–execution order, and raw-to-join energy equality are checked independently of the scheduling linear program (LP). Consequently, the flow witness certifies the exact committed ledger and measured profile, while the digest does not make an unverified claim about jobs that might have been submitted before the commitment. Experiment 16 reports this provenance certificate and reconciles the measured regional capacity envelope with the 118-MW nameplate that was committed before the validation/test split; the envelope is not used to select that nameplate.

The trust boundary is explicit. The verifier assumes that the scheduler and DCGM services are the authoritative sources for the fields used in ℓ_j , and that the digest is committed before the event decision. Under this assumption, a changed record is detectable and a joined record cannot be silently replaced after commitment. The digest does not, however, establish that a semantically false deadline, region label, service class, or energy value was truthful when it was signed. Authenticity of those fields requires an upstream scheduler attestation, meter signature, or trusted execution interface; it is outside the optimization certificate proved here.

B. Counterfactual, response, and false credit

For event set $\mathcal{E} \subset \mathcal{T}$, p_{dt}^0 denotes the no-event counterfactual and p_{dt}^1 denotes the event trajectory generated by the declared workload optimization. The mechanism-isolation experiments use $p_{dt}^{1, \text{sim}}$ for this optimized response, whereas Experiment 13 uses $p_{dt}^{1, \text{meas}}$ for the independently measured MIT/DCGM execution trace. We reserve p_{dt}^{obs} for measured execution used only as scoring truth; it is never supplied to an estimator or used to claim a causal utility event. For either response trace, $r_{dt} = p_{dt}^0 - p_{dt}^1$ is signed: gross local response is $[r_{dt}]_+$, while net response retains negative rebound. For an estimate $\hat{p}^0$, unsupported credited energy is

$$F = \Delta t \sum_{d,t \in \mathcal{E}} [\hat{p}_{dt}^0 - \max\{p_{dt}^{\text{obs}}, p_{dt}^1\}]_+. \quad (7)$$

The audit computes (7) separately for every locked day so that a favorable aggregate error cannot hide unsupported positive credit. The gross forecast credit is not itself a transfer. Once the event meter is closed, the payable response is the pointwise intersection

$$Q^{\text{pay}} = \Delta t \sum_{d,t \in \mathcal{E}} \min\{[\hat{p}_{dt}^0 - p_{dt}^{\text{obs}}]_+, [p_{dt}^0 - p_{dt}^{\text{obs}}]_+\}. \quad (8)$$

Equation (8) is a settlement rule, not an oracle-assisted forecast: the first term is the submitted gross credit and the second term is the response supported by the closed event meter. In the locked audit, p^0 is read only after the event as the trace-anchored no-event scoring profile; production deployment substitutes a precommitted no-event contract profile. Neither profile enters target fitting, gate decisions, or workload optimization. We report both quantities, as well as normalized root-mean-square error (nRMSE), bias, precision, recall, F_1 , under-credit, payment error, and overpayment.

C. Strategic reference scheduling

Consider an arithmetic mean of $H = 10$ reference days, event probability q , and DR price π^{DR} . If reference day j adds event-window service δR_j , the expected payment change is

$$\Delta\Pi = \frac{q\pi^{\text{DR}}}{H} \sum_{j=1}^H \delta R_j - \sum_{j=1}^H \Delta C_j. \quad (9)$$

Let c'_j be the right derivative of the minimum operating cost with respect to an event-window minimum-service constraint on day j . Linear-program sensitivity [18] gives the following exact local condition:

$$\frac{q\pi^{\text{DR}}}{H} > \min_j c'_j \iff \exists \text{ a strictly profitable local deviation.} \quad (10)$$

The historical-baseline premise is documented in [2]; (9)–(10) are this paper's workload-specific specialization.

D. Network value

Let M be the data-center-to-bus incidence matrix and $p_t = p_t^{\text{native}} + Mp_t^{\text{dc}}$ the nodal demand vector. For generator-to-bus map C_g , generator bounds, power transfer distribution factor (PTDF) H , and line ratings $\bar{f}$, the piecewise-linear security-constrained economic dispatch (SCED) value is

$$\begin{aligned} V(p) = \min_g \sum_i C_i(g_i) \\ \text{s.t. } \mathbf{1}^\top g = \mathbf{1}^\top p, \quad \underline{g} \leq g \leq \bar{g}, \\ -\bar{f} \leq H(C_g g - p) \leq \bar{f}. \end{aligned} \quad (11)$$

The incidence mapping aggregates device demand at its connection bus, and (11) is the standard lossless DC representation [14], [15]. The realized signed grid value is

$$\Delta V = V(p^0) - V(p^1). \quad (12)$$

Positive ΔV is a credit and negative ΔV a debit; no positive-only truncation is applied.

III. PROPOSED METHODOLOGY

A. Model-based framework and matched-information pre-estimation

The state, power, and network equations in Section II are the first stage of the methodology: they define the feasible workload state before any statistical target or payment is formed. Fig. 1 shows the complete chain. Meter history and the submitted-job ledger enter a statistical pre-estimator and the workload feasible set in parallel. The pre-estimator never receives execution energy used as scoring truth. For each event day, it uses only the information available under one of two declared protocols: lagged meter/calendar features for online baselines, or the complete submitted ledger for equally informed post-event comparisons. Ridge regression and gradient boosting are used as standard matched-information comparators [19], [20]; extremely randomized trees and synthetic control provide two additional nonlinear and donor-pool comparators [21], [22]. Two stronger comparators are also solved: median-loss gradient boosting with the complete submitted ledger and a nonnegative synthetic control whose weights globally minimize pre-event absolute error over 28 donor days. To separate the effect of projection from that of pre-estimation, the complete-ledger quantile output is also passed through the same exact workload-feasible projection, with its penalty chosen by contiguous validation folds. The same complete ledger is supplied to the post-event learner, the selected single projection, and the proposed convex verifier.

B. Sparse cumulative workload-state verifier

For statistical estimate $\tilde{p}$ and each predeclared penalty ρ_ℓ , the first-stage program solves

$$\min_{x,y,z} C_w(x) + \rho_\ell \sum_{d,t} z_{dt} \quad \text{s.t.} \quad z_{dt} \geq \pm(p_{dt}(x) - \tilde{p}_{dt}), \quad (13)$$

subject to (1)–(4). Absolute-value epigraphs are standard convex constructions [18]; using them to project a meter baseline onto an auditable workload ledger is specific to this work. Here C_w denotes workload operating cost; C_i denotes generator-segment cost in (11), and $V(p)$ denotes the network dispatch value. Thus (13) changes the objective, not the physical feasible set. Six globally solved schedules $p^{(\ell)}$ share identical arrivals and constraints. Their validation-only convex combination $\bar{p} = \sum_\ell \alpha_\ell p^{(\ell)}$, $\alpha \geq 0$, $\mathbf{1}^\top \alpha = 1$, remains feasible because the common scheduling set is convex.

The coefficients minimize validation squared error subject to both total and daily-tail false-credit budgets, where conditional value at risk (CVaR) measures the latter [23]. Let ℓ^* be the single candidate with minimum worst contiguous-fold nRMSE and B_j^{cap} its validation false exposure on day j . For reserve $\beta \in (0, 1]$,

$$\sum_j F_j(\alpha) \leq \beta \sum_j B_j^{\text{cap}}, \quad (14)$$

$$\text{CVaR}_{0.75}(F_j(\alpha)) \leq \beta \text{CVaR}_{0.75}(B_j^{\text{cap}}). \quad (15)$$

For implementation, the positive part is not differentiated through a sort. For each event sample n and validation day j , the program introduces $f_n \geq 0$, a CVaR threshold $\nu \geq 0$, and tail slacks $u_j \geq 0$:

$$f_n \geq p_n^\alpha - \max\{p_n^0, p_n^1\}, \quad F_j = \sum_{n \in j} f_n, \quad (16)$$

$$F_j - \nu - u_j \leq 0, \quad (17)$$

$$\nu + \frac{1}{0.25J} \sum_j u_j \leq \beta \text{CVaR}_{0.75}(B_j^{\text{cap}}). \quad (18)$$

The explicit epigraph in (18) is the implementation used in all reported fits. Together with the simplex equality, box bounds on α , and the total budget in (14), these are linear constraints and the squared error objective is a convex quadratic program. A HiGHS feasibility solve is followed by a sparse trust-region solve; the reported Karush–Kuhn–Tucker (KKT) and primal residuals are below 10^{-8} in every fitted fold. Four contiguous folds select β using validation nRMSE, then risk ratio; held-out noninferiority is reported as an outcome and is not used as a hidden constraint. A second exact projection targets $\bar{p}$ while imposing, for every event sample,

$$p_{dt}^{\text{safe}} \leq p_{dt}^{(\ell^*)}. \quad (19)$$

a) *Two-sided credit band.*: The upper envelope alone would admit an arbitrarily conservative zero-credit profile. We therefore impose the matching lower band on the event slots,

$$p_{dt}^{(\ell^*)} - \varepsilon \leq p_{dt}^{\text{safe}} \leq p_{dt}^{(\ell^*)}, \quad (d, t) \in \mathcal{D} \times \mathcal{E}, \quad (20)$$

where $\varepsilon = 1$ MW is fixed before the locked test split. The lower bound is implemented together with the workload constraints, not as a post-processing clip. Thus the output remains an attainable schedule while its credited event trajectory is within a declared band of the selected single projection. The total constraint (14) controls aggregate exposure and (15) controls the worst daily tail. The pair (19)–(20) is a deterministic cap relative to the selected workload-feasible reference; it is not a distribution-free guarantee against an arbitrary unseen meter trace. Overpayment against the realized event is handled by the

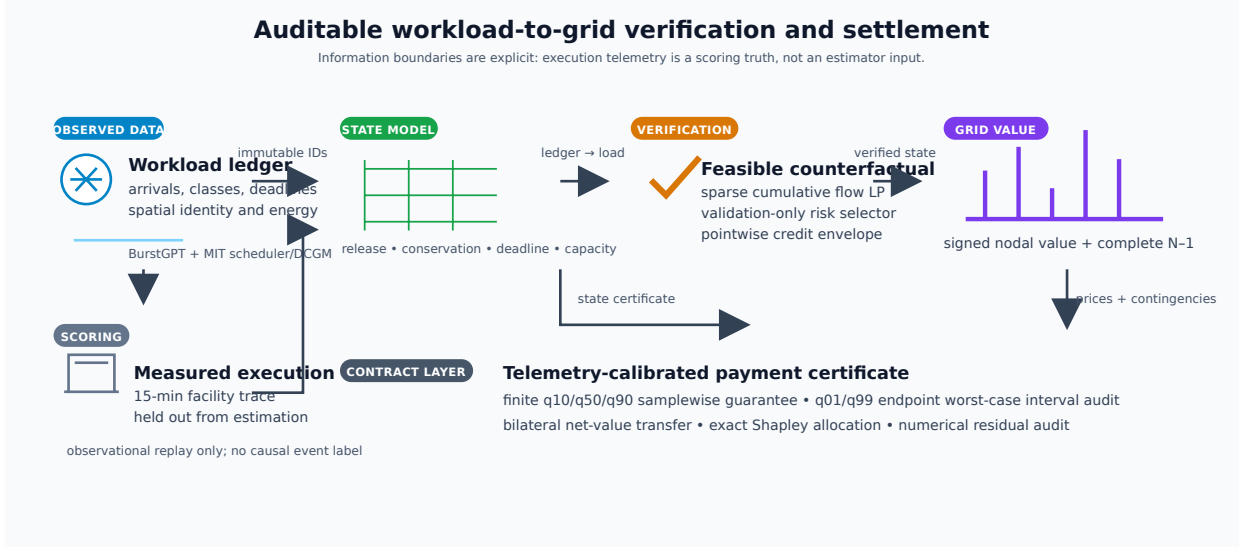


Fig. 1. Verification-to-settlement framework. Validation labels select the counterfactual risk contract; locked execution truth is used only for scoring. The right branch computes aggregate N-1 value and exact participant shares.

meter-capped rule (8). The selected single feasible projection proves nonemptiness, and the locked results report both margins and the post-event payable amount.

b) Decision-time information boundary. The ex-post selector can use the complete submitted ledger because it is an offline settlement protocol. For an event-gate deployment, all arrivals after the gate are removed before both the causal target construction and optimization, and terminal completion is required only for work due by the gate. The target is refit with the same metadata gradient-boosting specification used by the matched-information baseline, using historical days unchanged and the current day's masked ledger for every feature row. The committed-ledger rolling-service mode then projects this target over the physical fixed-plus-flexible capacity bound using only the masked ledger. It is an exact workload LP and never credits a reserved future job. After the event, its gross forecast credit is converted to the payable quantity in (8); the resulting nonzero payable response is therefore not obtained by suppressing the service trajectory or by using future arrivals. The rolling-service mode is reported separately from the information-boundary diagnostic and the complete-ledger benchmark.

To avoid conflating capacity planning with payment eligibility, the gate also publishes a causal reserve envelope. For a candidate quantile η and event day d , the reserve for a post-gate release cell is defined by (21):

$$\hat{a}_{sk\tau}^{\text{res}}(d; \eta) = Q_{\eta}(\{a_{sk\tau}(h) : h < d\}), \quad \tau \geq t_{\text{gate}}, \quad (21)$$

where the quantile is computed from temporally earlier days only. The finite candidate set is selected on validation days under a declared false-credit budget, and the resulting reserve is used for capacity planning only. The payable event profile remains the committed-ledger LP above; no reserved job can enter the payment numerator before its later ledger commitment. This separation supplies a causal forecast without silently reintroducing future arrivals into the certificate.

The cumulative state reduces the constraint-matrix nonzeros from $O(|S||K||D||T|^2)$ for explicit prefixes to $O(|S||K||D||T|)$. A 512-slot deadline is handled by (5) with every observed future arrival retained; no zero-arrival completion buffer is used in the final continuous-horizon certificate. The day trajectory is embedded by a lexicographic pair of linear programs: minimum L1 distance first, followed by minimum

operating cost on that optimal-distance face. Hence no penalty weight trades physical fit against cost. All linear programs are solved to global optimality with HiGHS [24]. The resulting state-to-credit sequence is summarized in Fig. 2; the execution replay and interval audit are reported separately in Experiments 13–15.

C. Signed nodal and N-1 settlement

Let $\lambda^0 \in \partial V(p^0)$ be the nodal demand-value subgradient. Signed linear settlement is $(\lambda^0)^{\top}(p^0 - p^1)$, while (12) is the realized system-value benchmark and the payoff of an explicit bilateral avoided-cost contract; it is not relabeled as an independent system operator (ISO) market payment. The convex subgradient inequality [18] yields the global certificate $0 \leq (\lambda^0)^{\top}(p^0 - p^1) - \Delta V = V(p^1) - V(p^0) - (\lambda^0)^{\top}(p^1 - p^0)$. (22)

Thus the linear rule globally upper-bounds exact value in the implemented polyhedral market model. Equality holds only when both endpoints share a supporting dual subgradient.

For market-compatible remuneration, let $\mathcal{W}$ contain the complete response cycle from the earliest deadline-feasible anticipatory shift through the latest recovery interval. The space-time rule is

$$\Pi^{\text{ST}} = \Delta t \sum_{t \in \mathcal{W}} \sum_d \lambda_{dt}^0 (p_{dt}^0 - p_{dt}^1). \quad (23)$$

This is the nodal virtual-link remuneration structure of [10], [11], applied here to a verified counterfactual. Summing site payments reproduces (23) exactly.

Space-time energy remuneration is only one component of the contracted DR product. Let $\mathcal{D}^{\text{part}}$ be the predeclared participating sites, $\mathcal{E}$ the event intervals, and r^{DR} the cleared service price. The signed delivered capacity service and total operator value are

$$Q^{\text{cap}} = \Delta t \sum_{d \in \mathcal{D}^{\text{part}}} \sum_{t \in \mathcal{E}} (p_{dt}^0 - p_{dt}^1), \quad (24)$$

$$G = r^{\text{DR}} Q^{\text{cap}} + \Pi^{\text{ST}}, \quad (25)$$

where p^0 is the minimum-cost no-event schedule under the identical continuous arrival window. Capacity credits and ex-post participation constraints follow established DR mechanism

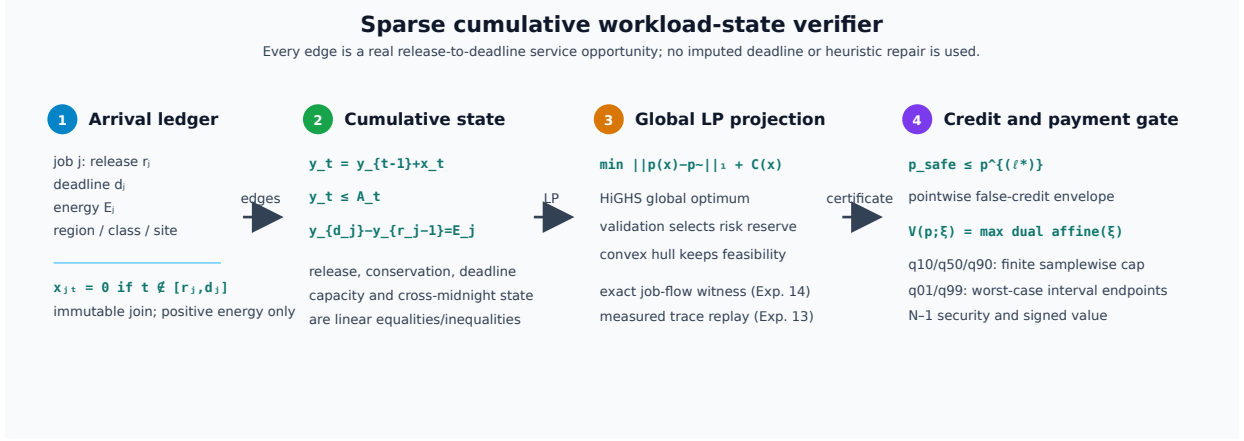


Fig. 2. Sparse cumulative workload-state verifier. Real release-to-deadline edges feed the global feasibility LP; the pointwise credit envelope and the finite/interval payment gates are applied only after the workload state is certified.

design [25], [26]; the signed Π^{ST} term separately debits spatial transfer and recovery. Equations (24)–(25) jointly define the signed service and operator value.

Let x^1 and x^0 denote the workload service schedules inducing p^1 and p^0 , respectively. Define $C_{\text{opp}} = C_w(x^1) - C_w(x^0)$ as the participant opportunity cost and $S = G - C_{\text{opp}}$ as the transferable transaction surplus. With disagreement utilities equal to zero, the symmetric Nash solution [27] activates the contract iff $S \geq 0$ and sets

$$z = \mathbb{I}\{S \geq 0\}, \quad P^B = z \left(C_{\text{opp}} + \frac{S}{2} \right), \quad u_{\text{dc}} = u_{\text{op}} = z \frac{S}{2}. \quad (26)$$

Thus (26) makes voluntary nonactivation the outside option rather than an omitted outcome. For every activated contract, both utilities are nonnegative and the single transfer P^B is exactly budget balanced. The bilateral transfer is reported separately from both ISO nodal remuneration and the exact dispatch value benchmark. Unlike event-only accounting, $\mathcal{W}$ debits both pre-event load increases and delayed rebound; workload conservation gives $\sum_{d,t \in \mathcal{W}} (p_{dt}^0 - p_{dt}^1) \Delta t = 0$ up to solver tolerance.

Security-aware settlement augments (11) with every finite non-islanding single-line outage. If $L_{\ell k}$ is the LODF from outage k to monitored line ℓ , then

$$-\bar{f}_\ell \leq (H_\ell + L_{\ell k} H_k)(C_g g - p) \leq \bar{f}_\ell, \quad \ell \neq k. \quad (27)$$

Equation (27) is the standard preventive N-1 LODF formulation [16]; the experiment enforces all credible rows simultaneously, rather than screening contingencies. Native public ratings are retained. Islanding outages are reported and excluded because a connected single-reference PTDF cannot represent separate islands, consistent with the scope of DC security models and the N-1 planning criterion [28].

D. Scenario-robust N-1 payment certificate

The pointwise envelope controls credited energy but does not by itself certify network payment because spatially different demand vectors can activate different security constraints. Let $\mathcal{P} = \text{conv}\{p^{(1)}, \dots, p^{(L)}\}$ be the convex hull of complete workload-feasible schedules and let $p^{\text{cap}} \in \mathcal{P}$ be the validation-selected single feasible projection. Its penalty ρ_{ℓ^*} is selected by contiguous validation and is reported separately from the payment-target penalty. The feasible-quantile projection is retained as an external matched transfer comparator in

Section V. Let Ξ_3 contain the 10th, 50th, and 90th percentiles of held-out per-job measured-to-predicted GPU energy. These predeclared empirical scenarios define the finite power-conversion contract supported by independent telemetry [29]. For each settlement day, the payment-certified verifier defines $B_\xi^{\text{cap}} = \Delta t \sum_{t \in \mathcal{E}} V_{N-1}(p_t^{\text{native}} + M \xi p_t^{\text{cap}})$ and solves

$$\begin{aligned} & \text{lexmin}_{\alpha, e, \eta, \{g_t^\xi\}} \left(\sum_{d, t \in \mathcal{E}} e_{dt}, -\eta \right) \\ & \text{s.t.} \quad p^\alpha = \sum_{\ell=1}^L \alpha_\ell p^{(\ell)}, \\ & \quad \alpha \geq 0, \quad \mathbf{1}^\top \alpha = 1, \quad 0 \leq \eta \leq 1, \\ & \quad e_{dt} \geq \pm(p_{dt}^\alpha - p_{dt}^{\text{safe}}), \\ & \quad g_t^\xi \text{ satisfies (11) and (27),} \\ & \quad \text{with } p_t = p_t^{\text{native}} + M \xi p_t^\alpha, \\ & \quad \quad \quad t \in \mathcal{E}, \quad \xi \in \Xi_3, \\ & \quad \Delta t \sum_{t \in \mathcal{E}} \sum_i C_i(g_{it}^\xi) \\ & \quad \leq (1 - \eta) B_\xi^{\text{cap}}, \quad \xi \in \Xi_3. \end{aligned} \quad (28)$$

Generator costs use the same declared piecewise-linear epigraph as market settlement. Two global linear programs implement the lexicographic objective: the first obtains the minimum trajectory error, and the second preserves that optimum within solver tolerance while maximizing the common fractional margin η . Existence of the primal dispatches in (28) is equivalent to the corresponding upper bound on the optimal N-1 value; the unit weight on p^{cap} proves nonemptiness in every scenario. Consequently, for any $\xi \in \Xi_3$ and realized event load $p^{1,\xi}$, daily payment obeys

$$\begin{aligned} & \sum_{t \in \mathcal{E}} \Delta t [V_{N-1}(p_t^{\text{native}} + M \xi p_t^\alpha) - V_{N-1}(p_t^{1,\xi})] \\ & \leq \sum_{t \in \mathcal{E}} \Delta t [V_{N-1}(p_t^{\text{native}} + M \xi p_t^{\text{cap}}) - V_{N-1}(p_t^{1,\xi})]. \end{aligned} \quad (29)$$

The common realized-cost term cancels, so the certificate requires no execution truth and holds samplewise. Both programs contain every finite non-islanding contingency. The formulation follows the standard SCED primal and linear-program epigraph foundations [15], [18]. Here $\mathcal{P}$ contains the six independently solved first-stage projections from (13) and

the matched feasible-quantile projection. The risk-constrained verifier is an external target in the absolute-deviation objective and is not a selectable candidate; the quantile schedule is used only as an independent transfer comparator, not as the contractual payment cap.

The finite cap is complemented by a set-valued payment certificate. The contractual cap profile is denoted $p^{\text{cap}} = p^{(\ell^*)}$, where ℓ^* is selected by contiguous validation nRMSE; the independent payment-target profile p^{pay} is selected separately by validation payment error and is used only as the target of the certificate. For a realized counterfactual value, define $\Delta V_\xi(p) = V_{N-1}(p, \xi) - V_{N-1}(p^{1:\xi})$ and the continuous segment $p_\theta = (1 - \theta)p^{\text{cap}} + \theta p^{\text{cert}}$, $0 \leq \theta \leq 1$. The induced interval is

$$\mathcal{I}_\xi = \left[\min_{0 \leq \theta \leq 1} \Delta V_\xi(p_\theta), \max\{\Delta V_\xi(p^{\text{cap}}), \Delta V_\xi(p^{\text{cert}})\} \right]. \quad (30)$$

The lower endpoint is obtained by one exact joint N-1 SCED LP with a shared segment parameter; the upper endpoint follows from convexity of the optimal dispatch value. The selected cap payment is an element of $\mathcal{I}_\xi$. The endpoints use only validation-frozen feasible profiles and the realized counterfactual needed for settlement; no oracle baseline selects or certifies the interval. Interval width is reported as uncertainty cost, while the seven-profile hull remains the finite-scenario payment certificate in (28).

E. Participant allocation

For participants N , define $v(S)$ as signed dispatch value when coalition S moves from baseline to actual demand and nonmembers remain at baseline. The exact allocation is

$$\phi_i(v) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|!(|N| - |S| - 1)!}{|N|!} [v(S \cup \{i\}) - v(S)]. \quad (31)$$

This is the Shapley value [30]; the characteristic function based on signed grid value is the proposed specialization. Grouping (31) by coalition size cancels every intermediate value and gives $\sum_i \phi_i = v(N)$ when $v(\emptyset) = 0$. We enumerate all 16 coalitions for four sites and all 256 coalitions for eight contractual portfolios; no permutation estimator is used. For larger auditable contracts, exchangeable slices at the same electrical site are grouped by their member count. The exact sum retains the binomial multiplicity of every labeled coalition and evaluates only $\prod_g (m_g + 1)$ distinct count states. This symmetry reduction remains the Shapley value and is evaluated through 20 participants.

F. Analytical guarantees

Proposition 1 (manipulation boundary): Under a ten-day arithmetic baseline, differentiable one-sided optimal costs, and an interior feasible increase in at least one reference-day event service, a strictly profitable local manipulation exists if and only if (10) holds. To prove the statement, differentiate (9) along any feasible reference-day direction. The marginal payment is the same $q\pi^{\text{DR}}/H$ for every day, whereas the least physical cost is $\min_j c'_j$. A strict excess selects a positive direction on an argmin day; if no strict excess exists, every nonnegative local direction has nonpositive first-order profit. Equality admits indifference but does not imply strict profit. Experiment 1 computes c'_j from the honest program's right-hand-side dual and solves the strategic program independently, preventing the behavioral result from being reused as its own certificate.

Proposition 2 (workload and two-sided credit feasibility): For fixed arrivals, every convex combination of solutions to (1)–(4)

is feasible, and the envelope output satisfies the deterministic two-sided band (20) on every event cell. Consequently, for every possible unseen execution c_{dt} , its false-credit energy is no larger than that of $p^{(\ell^*)}$, while its additional under-credit is bounded by $\varepsilon|\mathcal{D}||\mathcal{E}|\Delta t$ relative to that reference. Affine equalities are preserved by the simplex coefficients; each linear inequality is preserved by convexity. The upper inequality in (20) and monotonicity of $[z - c]_+$ give

$$[p_{dt}^{\text{safe}} - c_{dt}]_+ \leq [p_{dt}^{(\ell^*)} - c_{dt}]_+.$$

The lower inequality gives $[c_{dt} - p_{dt}^{\text{safe}}]_+ \leq [c_{dt} - p_{dt}^{(\ell^*)}]_+ + \varepsilon$. Summation over sites and event intervals proves both bounds. The result is a feasibility-and-band certificate determined by the declared constraints and ε , not by the distribution of the locked test set.

Proposition 3 (security-aware value consistency): For the piecewise-linear base-case or N-1 dispatch, exact signed settlement equals the corresponding optimal-value difference and the nodal linear rule upper-bounds it by (22). The feasible-set right-hand side is affine in nodal demand, so the optimal value is convex and piecewise affine. Applying the subgradient inequality at p^0 gives (22); applying it again at p^1 yields

$$\begin{aligned} 0 \leq \Delta V_{\text{lin}} - \Delta V &\leq (\lambda^1 - \lambda^0)^\top (p^1 - p^0) \\ &\leq \|\lambda^1 - \lambda^0\|_* \|p^1 - p^0\|. \end{aligned}$$

The same proof applies after stacking all rows in (27), because they remain linear. Gross positive-only settlement has no analogous identity: truncation changes the response vector before it enters the value function.

Proposition 4 (robust daily payment noninferiority): Every solution of (28) is workload feasible and its exact N-1 payment is no larger than the contractual cap payment for every realized event trajectory and every declared $\xi \in \Xi_3$. Workload feasibility follows from Proposition 2. For each ξ , the embedded dispatch is a feasible witness whose cost satisfies the corresponding last constraint in (28); optimal N-1 dispatch can only have lower cost. Subtracting the same scenario-specific realized cost gives (29). Conditional on the telemetry-derived finite set Ξ_3 , the proof is deterministic and contains no test-outcome assumption.

Proposition 5 (continuous worst-case interval certificate): For a fixed workload profile p , the optimal piecewise-linear N-1 dispatch value $V_{N-1}(p; \xi)$ is convex in a scalar conversion factor $\xi \in [\xi_L, \xi_U]$. Therefore

$$\max_{\xi \in [\xi_L, \xi_U]} V_{N-1}(p; \xi) = \max\{V_{N-1}(p; \xi_L), V_{N-1}(p; \xi_U)\}.$$

The value function is the supremum of affine dual objectives in ξ , hence convex; a convex function on a compact interval attains its maximum at an endpoint. If the two endpoint inequalities $V_{N-1}(p^\alpha; \xi_e) \leq V_{N-1}(p^{\text{cap}}; \xi_e)$ hold for $e \in \{L, U\}$, then the certified candidate also has no larger worst-case interval cost than the contractual cap. This interval result is a worst-case bound; pointwise dominance at every interior ξ is asserted only for the finite set Ξ_3 in Proposition 4. Experiment 15 evaluates the capacity-safe endpoints with the exact four-segment contractual N-1 model, while Experiment 9 independently replays Ξ_3 at 40 segments; the two resolutions are not conflated.

Proposition 6 (workload-segment payment interval): For a fixed conversion factor and realized counterfactual, the lower endpoint in (30) is the exact minimum payment over the declared continuous segment because the shared segment parameter and all interval-wise preventive N-1 dispatch variables are solved in one joint LP. The upper endpoint is the exact maximum because the optimal dispatch value is convex in the affine profile and a convex function attains its maximum on a compact segment at a vertex. Thus (30) is exact for the contractual segment,

TABLE I
COMPLETE DATA AND EVALUATION PROTOCOL.

Item	Declared use and count
BurstGPT	5,188,507 requests; 2.176 billion tokens; inference arrivals and observed service
MIT scheduler	287,173 raw rows; 104 duplicate rows removed, leaving 287,069 canonical rows; submission time, duration, GPU count, and queue state
MIT GPU telemetry	96,893 rows; 68,664 tensor-window jobs and 71,128 full-horizon positive-energy jobs; measured execution energy
Time resolution	15 min; 121 calendar days; 121 valid complete days
Evaluation	16 validation days followed by 54 untouched test days
Networks	IEEE Reliability Test System 24-bus (RTS-24), IEEE-30, IEEE-39, and IEEE-118 for AC/N-1 panels; IEEE-300 is added in the resolution-convergence panel
Provenance	Immutable scheduler/DCGM join digest, raw-to-join energy conservation, and measured regional capacity envelope

not a grid approximation or a claim that the segment contains every physically possible schedule. Experiment 15 reports the joint-LP minimum, endpoint maximum, and resulting width at both held-out conversion endpoints; no oracle baseline selects the segment.

Proposition 7 (provenance-bound workload witness): Suppose the canonical tuple in (6) is computed from the retained scheduler–DCGM join after the declared duplicate and positive-energy rules. If the certificate verifies one-to-one joining, release–execution ordering, raw-to-join energy conservation, and zero LP residuals, then the job-level flow witness is a feasible representation of that committed ledger. Under the collision-resistance assumption of SHA-256, any post-commit modification of a tuple or its canonical ordering changes the digest except with negligible probability. The statement binds the computational witness to the committed records; it does not certify that an uncommitted pre-event submission was truthful. Experiment 16 checks each premise independently and records the raw source hashes, so the provenance claim is separate from the scheduling solver’s own feasibility status.

IV. EXPERIMENTAL DESIGN

A. Data, preprocessing, and information boundary

Table I reports the complete data inventory. BurstGPT contributes 5,188,507 successful requests and observed token service [31]. MIT Supercloud contributes scheduler submissions and independent Data Center GPU Manager energy. The complete positive-energy scheduler–DCGM join contains 71,128 jobs, of which 68,664 start within the common 121-day tensor window used by Experiments 1–13 [29]. Submission time creates queue arrivals; measured execution intervals create scoring truth. A held-out GPU-power regression uses 28,413 observations and obtains $R^2 = 0.899$, mean absolute error (MAE) 11.22 W, and root-mean-square error (RMSE) 15.82 W. All 99th-percentile workload scaling factors are fitted on the first 40 complete days of the training portion; validation and locked-test days are excluded from that fit. Per-job measured-to-predicted energy ratios define Ξ_3 in (28); no synthetic row replaces an invalid record. The 68,664 count is the 121-day workload tensor used by Experiments 1–13. Experiment 14 does not truncate at that tensor boundary: it extends the deadline horizon to the last observed scheduler completion and retains all 71,128 positive-energy joined jobs for its independent job-level flow witness.

The public traces do not contain utility bus locations. Consequently, the four-region coupling is a trace-driven benchmark

with fixed, predeclared bus placements and a declared workload-to-power conversion, not a utility field trial. Immutable row identifiers determine spatial partitions before any outcome is computed. Held-out measured-to-predicted job-energy ratios span 0.5845–2.6931 (1st–99th percentiles); the contractual interval uses the capacity-safe upper factor 1.1499 because the raw 99th-percentile scale would exceed the precommitted 118-MW nameplate. Experiments 9, 11, and 15 retain the unclipped factor as a calibration stress value rather than treating the 118-MW scale as a measured geographic fact. This design permits exact repeated mechanism comparisons but does not identify a particular operator’s geography. Public PGLib and MATPOWER/PYPOWER cases provide topology, capacities, costs, and ratings [15], [32]. The RTS-24 benchmark follows the published system data [33].

B. Baselines, parameters, and inference

Experiment 2 compares High-5-of-10, ridge regression, gradient boosting, Extra Trees, online metadata gradient boosting, equally informed post-event metadata gradient boosting, complete-ledger quantile boosting, pre-event synthetic control, the exact feasible projection of the complete-ledger quantile, the tail-risk counterfactual, the independently selected single feasible projection, and the risk-constrained verifier. The first eight are meter/statistical comparators; the remaining methods receive the declared ledger as appropriate. The payment-certified verifier is evaluated separately in Experiments 9 and 15 because its N-1 payment certificate uses a different objective and a frozen candidate hull. An information-set audit verifies feature equality and confirms that no method observes execution truth.

Real-time inference has zero deferral, elastic inference four slots, and batch GPU work 512 slots. Per-site flexible capacity is 118 MW plus 6 MW fixed facility demand; the 118-MW nameplate is committed before the validation/test split and Experiment 16 only reconciles it with the complete observed MIT/DCGM batch envelope. Six projection penalties $\{0.03, 0.1, 0.3, 1, 3, 10\}$ and four risk reserves $\{0.60, 0.75, 0.90, 1.00\}$ are selected only on validation data. The settlement uses 10 generator-cost segments and is scored by an independently solved 40- or 80-segment evaluator. All 5,000 bootstrap replicates use moving three-day blocks [34]. Paired tests enumerate all 2^{18} signs of 18 nonoverlapping three-day blocks; Holm correction controls family-wise error [35].

The sparse implementation grows from 5,760 variables and 13,424 nonzeros at 96 slots to 36,480 variables and 85,488 nonzeros at 608 slots; the recorded HiGHS solve times are 0.039–0.212 s on the same host. This scaling makes the deployment-time computational burden auditable rather than implied by the asymptotic claim alone.

The nonlinear model-out check uses the standard polar AC optimal power flow (OPF) with active/reactive balance, voltage bounds, generator reactive bounds, and apparent-power ratings [15]. After solving the intact state, the preventive panel imposes, for every non-reference generator i and every credible outage k ,

$$P_{Gi}^{(k)} = P_{Gi}^{(0)}, \quad i \notin \mathcal{G}_{\text{ref}}, \quad k \in \mathcal{C}. \quad (32)$$

The reference generator balances only outage-dependent active losses; $Q_G^{(k)}$ and bus voltages are contingency recourse. Thus (32) tests one shared active plan rather than separately optimized post-contingency active schedules.

Experiments 1–2 test incentives and workload verification; 3–9 test signed settlement, network transfer, risk certification, and

N-1 value; 10-12 test AC feasibility, spatial/scale transfer, and continuous-cycle remuneration. Experiments 13-16 provide measured replay, a job-level witness, conversion intervals, and immutable ledger reconciliation. Experiment 17 freezes the event-gate information set, and Experiment 18 evaluates AC post-contingency recourse across four public networks. Confirmatory comparisons use the same 54 locked days unless a deterministic network rule is stated.

V. RESULTS

A. Incentive boundary and independent verification

Experiment 1 solves the entire 8×7 DR-price/call-probability grid. The dual threshold (10) agrees with the independently solved strategic program in all 56 cells. At \$150/MWh and $q = 0.35$, strategic reference scheduling inflates the event baseline by 4.09 MWh. Of 52.46 MWh credited, 18.05 MWh is unsupported by physical reduction, giving a false-credit ratio of 0.344. The profitable region follows the derived marginal boundary rather than a fitted classifier; the complete grid and its phase diagram are retained in the experiment output.

Experiment 2 evaluates all twelve methods on 54 later days. Table II separates the high-recall tail-risk counterfactual from the settlement-safe output. The tail-risk counterfactual improves mean credit F_1 from 0.441558 to 0.484219 over the selected single projection (Holm-adjusted $p = 0.000069$); its nRMSE is 0.330795 versus 0.345209, and the block test does not reject equality ($p = 0.130165$). The risk-constrained verifier combines seven independently feasible candidates without degenerating to one candidate. Relative to the equally informed feasible-quantile projection, unsupported credit falls from 2.917471 to 1.826322 MWh/day, a 37.40% reduction (Holm-adjusted $p = 0.018494$), while nRMSE changes from 0.321370 to 0.345019 (Holm-adjusted $p = 0.546875$). The F_1 change is 0.508048 to 0.441220 (Holm-adjusted $p = 0.000053$), so the result is selective risk control rather than a claim of uniformly better classification. The moving three-day block 95% interval for the paired false-credit reduction against feasible quantile is [0.5474, 1.7660] MWh/day; against the selected single projection it is only [0.0054, 0.0465] MWh/day. These intervals make the magnitude and uncertainty of the two comparisons explicit. Against the selected single projection, RiskSafe changes false credit from 1.849211 to 1.826322 MWh/day, nRMSE from 0.345209 to 0.345019, and F_1 from 0.441558 to 0.441220; all three paired tests are nonsignificant. The matched ledger controls therefore attribute the principal effect to the feasible set and deterministic envelope, rather than execution truth.

The new risk fit is a convex quadratic program rather than a nonsmooth sorted-positive-part optimization: the selected reserve is $\beta = 1$ after four contiguous validation folds, and the final fit has KKT stationarity 9.4×10^{-10} and zero primal residual. On the locked test days, the pre-envelope total-plus-CVaR fit changes false credit by +0.674285 MWh/day and nRMSE by -0.014414 relative to the single reference; CVaR alone changes false credit by +1.949153 MWh/day, whereas the total budget alone changes it by +0.673563 MWh/day. The subsequent exact pointwise envelope changes false credit by -0.022889 MWh/day relative to that reference. This decomposition separates the convex risk fit from the pointwise certificate rather than attributing the latter's bound to CVaR. The two-sided certificate in (20) is checked independently for all 54 locked days and every event cell. With the predeclared 1-MW band, the final mean false-credit exposure is 1.826322 MWh/day and mean under-credit is 13.446065 MWh/day, compared with 13.430785 MWh/day for the selected single

TABLE II
LOCKED 54-DAY COUNTERFACTUAL RESULTS (DAILY MEANS).

Method	nRMSE	False MWh	Under-credit	F_1
High-5-of-10	4.214	172.709	6.833	0.144
Ridge	2.772	110.666	5.625	0.235
Gradient boosting	2.351	97.057	5.330	0.246
Extra Trees	3.336	132.971	6.947	0.177
Online metadata	2.133	90.589	4.663	0.264
Post-event metadata	2.124	91.603	4.559	0.267
Post-event quantile	0.944	24.156	6.618	0.532
Synthetic control	2.552	101.835	8.427	0.206
Feasible quantile	0.321	2.917	11.493	0.508
Single feasible	0.345	1.849	13.431	0.442
Tail-risk feasible	0.331	2.523	11.980	0.484
Risk-constrained safe	0.345	1.826	13.446	0.441

cap; every upper and lower pointwise margin is nonnegative. Relative to feasible quantile, RiskSafe reduces false credit by 37.40% but increases under-credit by 1.953 MWh/day and does not improve F_1 . It is therefore a certified risk tradeoff, not a claim of uniform predictive dominance.

The constraint-set ablation isolates the two risk components. Removing both the total-exposure and CVaR constraints yields 4.679510 MWh/day of false credit on the locked days; retaining only the total budget gives 2.522774, retaining only CVaR gives 3.798364, and retaining both gives 2.523496 MWh/day. The two budgets therefore have distinct effects: total exposure controls the aggregate false-credit level, whereas CVaR controls the tail without being a surrogate for the pointwise envelope. These values come from exact convex programs with the same candidate set and locked scoring trace; the full two-sided projection is audited separately above.

The closest domain baselines are evaluated on the same locked ledger, but are labelled structural analogues rather than claimed reimplementations of the cited algorithms. The incentive-compatible spatial-DR, flexibility, receding-horizon, and multi-site aggregator analogues have daily means (nRMSE, false credit, F_1) of (3.063, 114.572, 0.137), (0.370, 1.549, 0.411), (0.345, 1.849, 0.442), and (0.329, 0, 0.279), respectively. A second panel implements the recent DC-T and DC-ST workload formulations as exact ledger LPs with native-site assignment fixed or cross-site migration enabled; both have nRMSE 0.370104 and false credit 1.549206 MWh/day, while their mean migration is 0 and 4.327863 MWh, respectively. These controls are equation-level translations on the identical ledger, deadlines, prices, and capacities; they are not presented as reproductions of the published software implementations. The spatial incentive and flexibility formulations are represented by the first two controls [7], [26]; the receding-horizon and aggregator controls follow the latter two decompositions [8], [12]. The recent security-constrained spatio-temporal formulation is represented by the exact DC-T/DC-ST ledger panel, not by the four structural analogues [6]. This explicit separation prevents a structural proxy from being mistaken for a faithful modern literature baseline.

Fig. 3 reports the complete day-level distributions, constraint ablation, validation frontier, and independent intervention robustness. The 5-by-3 capacity/deadline factorial contains 810 optimal schedules; maximum capacity- and deadline-binding shares are 0.310 and 0.948. The measured nonzero effects establish that the associated constraints are not decorative. Matrix nonzeros scale with log-log slope 1.003 over 96-608 intervals, matching the linear-horizon proposition.

B. Nodal value and spatial response

Experiment 3 crosses thirteen counterfactuals with five settlement mechanisms. Table III first uses the trace-anchored

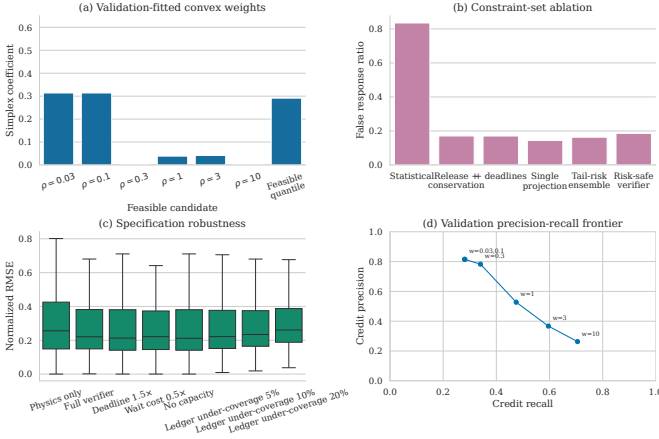


Fig. 3. Validation-fitted projection weights, constraint-set ablation, specification robustness, and the precision-recall frontier. Every point is produced by a globally solved workload program.

TABLE III
BASE-CASE AND COMPLETE N-1 SETTLEMENT ERRORS (USD/DAY).

Panel	Settlement	Trace MAE	Workload MAE	Trace median/overpay	Workload median/overpay
Base case	Uniform gross	555.923	501.511	—	—
Base case	Nodal gross	555.910	501.324	—	—
Base case	Uniform signed	3.956	454.116	—	—
Base case	Nodal signed linear	3.953	454.366	—	—
Base case	Nodal exact net value	2.230	454.238	—	—
N-1	Base exact	165.288	192.978	85.21/156.62	121.75/134.29
N-1	N-1 linear	2.253	159.514	0.59/1.97	115.96/38.96
N-1	N-1 exact	1.018	159.483	0.98/0.01	116.47/38.37

reference to isolate the mechanism. Exact signed nodal value has mean absolute error \$2.230, versus \$555.910 for nodal gross response. Under the workload verifier, the errors are \$454.238 and \$501.324, respectively; the remaining error is dominated by counterfactual uncertainty rather than value-function linearization. The paired one-factor decomposition prevents attributing this bundled improvement to location alone. With the trace reference, signed netting lowers error by \$551.957/day (Holm-adjusted exact block-sign $p = 8.0 \times 10^{-6}$), whereas changing uniform signed prices to nodal prices changes it by only \$0.0032/day ($p = 0.4726$), and replacing nodal linear value by exact value lowers it by \$1.723/day ($p = 0.2749$). Under the workload verifier, the corresponding effects are \$46.958, -\$0.250, and \$0.128/day; none is a significant universal improvement after the paired block correction. Thus signed rebound accounting identifies the dominant trace-layer gain, while end-to-end value differences remain modest and are tested separately in the congested and N-1 panels below. All 5,616 interval/counterfactual instances satisfy (22); the largest numerical violation is 2.03×10^{-11} dollars.

Experiment 4 applies a predeclared rule—the largest gross-minus-net spatial offset—and selects day 69. The source curtailment, remote rebound, nodal prices, and line utilization are retained in the experiment output. The case explains why positive-only local accounting fails: migration creates a simultaneous negative component that gross payment discards.

C. Cross-network and N-1 security evidence

Experiment 5 contains 6,480 outcomes over four public networks, three native-load multipliers, two counterfactual qualities, five mechanisms, and 54 days. Exact value improves trace-anchored error in all 12 network-load cells (mean \$2.126/day; eight Holm-adjusted tests below 0.05), whereas the workload-verifier reduction is only \$0.147/day (765.776 to 765.629) and none of its 12 tests is significant. The near-affine RTS-24 resolution panel changes signed-linear-minus-exact error by 0.010632–0.030576/day across four segment pairs, all

below 0.04, supporting the stated value geometry without an end-to-end dominance claim.

Experiment 6 crosses five capacity and three deadline multipliers over 54 days (810 globally solved schedules). Capacity- and deadline-binding shares reach 0.0334 and 0.9118; mean false-credit exposure across the capacity levels is 2.700, 2.407, 2.303, 2.297, and 2.407 MWh, while deadline binding remains 0.8738–0.9118. Every schedule satisfies the same release, deadline, and capacity constraints.

Experiment 7 enumerates all 16 four-site and 256 eight-site coalitions. The trace-reference Shapley error is zero; workload-verifier mean participant error is \$92.519, with maximum budget residuals 1.14×10^{-13} dollars and an exact 20-participant panel (1,296 states). Allocation is therefore enumerated, not permutation-sampled.

Experiment 8 enforces (27) for all 37 non-islanding RTS-24 outages over 2,592 locked interval-mechanism outcomes and scores the 10-segment certificate with an independent 40-segment dispatch. Trace-anchored errors for base exact, N-1 linear, and N-1 exact are \$165.288, \$2.253, and \$1.018/day; the latter improves over base exact ($p = 3.1 \times 10^{-5}$) but not over N-1 linear ($p = 0.3518$). Workload errors are \$192.978, \$159.514, and \$159.483 ($p = 0.3518$ and 0.9309), with maximum post-contingency utilization 1.000 p.u. The trace-anchored row is a calibration identity for the network-value layer, not an independent end-to-end prediction test; the workload rows use the locked counterfactual and the finer independent evaluator and therefore retain the substantially larger errors shown above.

Experiment 9 tests (29). All 54 daily lexicographic program pairs attain global optimality; the seven-profile hull and independent payment target are frozen on validation. The held-out factors 0.734, 0.996, and 1.278 form Ξ_3 ; all 162 certificates have zero positive cap violation and mean cost margins \$17.506, \$23.542, and \$30.346/day. Independent 40-segment replay gives mean absolute errors \$194.282, \$179.261, \$190.284, and \$179.272/day for payment-certified verifier, the selected single, feasible quantile, and risk-constrained verifier, respectively; these accuracy scores do not replace the deterministic cap. Maximum post-contingency loading is 1.000 p.u. Experiment 15 separately audits the capacity-safe closed conversion interval and is interpreted through Proposition 5. Its exact four-segment continuous-profile certificate has mean payment-interval width \$13.947662/day at q_{01} and \$27.210301/day at the capacity-safe q_{99} endpoint (maximum widths \$92.035916 and \$182.027759/day). The raw held-out $q_{99} = 2.6931$ is retained as an unclipped calibration value but exceeds the precommitted 118-MW nameplate; the contractual upper endpoint is therefore the ledger-audited safe factor 1.149889. This interval is an explicit uncertainty cost, not a lower-error claim. The cap is a contractual upper bound under the declared market model; it is not evidence that the payment-certified profile has lower independent prediction error than every comparator. The oracle payment is evaluated only after the interval is frozen as an independent coverage diagnostic; it is not used to select a profile, an endpoint, or a settlement cap. Only 8 of 108 oracle endpoint observations (7.4%) lie inside the closed interval, so the interval is a deterministic conversion-factor certificate, not a statistical prediction interval for arbitrary workload-model error. This coverage audit is the reason the conversion result is not used as evidence of probabilistic calibration.

Experiment 10 replaces the linear network with nonlinear AC OPF [15]. All 864 RTS-24/30/39/118 base outcomes converge with zero voltage violation and maximum loading 1.00000003 p.u. (excess 2.1×10^{-8} , below the 10^{-6} tolerance). The cor-

TABLE IV
EVENT-GATE INFORMATION BOUNDARY AND POST-EVENT SETTLEMENT
(54-DAY MEANS).

Method	nRMSE	Gross false MWh	Meter-capped MWh	Recall
Gate-truncated ledger	0.451	1.258	0.995	0.033
Committed rolling service	0.451	1.258	0.995	0.033
Committed-ledger reference	0.452	0	0	0
Complete-ledger RiskSafe	0.345	1.826	8.842	0.353

rective IEEE-9/14 panel has 5,400 converged OPFs, maximum loading 0.8816 p.u., and zero voltage violation; the shared-active-plan panel has 3,888 cells, zero non-reference active-plan deviation, and maximum loadings 0.878/0.903/0.929 p.u. at 3%/6%/9% penetration. The payment claim remains scoped to (28).

Experiment 11 crosses all $4! = 24$ region-to-bus mappings, 3%/6%/9% penetrations, four counterfactuals, and 54 days (15,552 outcomes). RiskSafe errors are \$100.220/201.509/296.951/day; feasible quantile is \$84.336/169.715/249.217, while RiskSafe is lower than the selected single (\$100.270/201.610/297.102) in all 72 mapping-scale cells. The panel thus tests spatial/scale transfer without relabelling the external quantile control. Experiment 16 audits 66,769 training and 28,413 test jobs (MAE 11.223 W, RMSE 15.818 W, $R^2 = 0.8994$, aggregate ratio 1.0644); the held-out q_{99} exceeds the 118-MW commitment and is clipped to 1.149889. The four-bus placement remains a declared scenario.

Experiment 12 embeds each event in a 1,216-slot horizon with 512 recovery slots. Across 216 outcomes, maximum window and cycle-energy residuals are 6.0×10^{-6} MW and 3.0×10^{-6} MWh; complete-cycle MAE is \$0.084761, \$0.053170, \$0.241926, and \$0.084761/event for payment-certified verifier, feasible quantile, risk-constrained verifier, and single projection. All 54 Nash contracts activate (mean surplus \$2,034.81; minimum utility \$1.84).

D. Independent trace and deployment-boundary audits

Immutable MIT scheduler/DCGM replay gives slot MAE 3.744 versus 5.764 MW for the batch-only trace with 100% coverage; the exact flow witness retains 71,128 jobs with residual below 10^{-17} MWh. Raw/scaled peaks are 0.006685/102.619 MW, and the $q_{01}/q_{99\text{-cap}}$ interval widths are \$13.948/27.210 per day with cap violation below 3×10^{-9} USD. The gate masks post-gate arrivals. The rolling profile has nRMSE/gross-false-credit/recall 0.451393/1.257628/0.032868 and 0.995 MWh/day meter-capped payable response; the complete-ledger RiskSafe comparator is 0.345019/1.826322/0.353429, while the reference is an eligibility control with no gross credit. Table IV gives the locked-day comparison. Its meter-capped column is post-event settlement; the zero false- payment entry is an accounting invariant, whereas gross false credit remains the independent risk metric. The selected $\eta = 0.60$ reserve is capacity planning only; 1,698 AC rows are model-boundary audits rather than a field event.

VI. CONCLUSION

This paper develops a provenance-bound workload-state verifier and security-aware nodal settlement. Across 54 locked days, RiskSafe has false credit 1.826322 MWh/day and nRMSE 0.345019 versus 1.849211/0.345209 for the single cap and 2.917471/0.321370 for feasible quantile; this is a certified risk tradeoff, not uniform predictive dominance. Trace-anchored N-1 exact settlement error falls from \$165.288 to \$1.018/day, while end-to-end errors are \$192.978/159.514/159.483 for base

exact/N-1 linear/N-1 exact. The job witness retains 71,128 measured jobs. At the event gate, the rolling committed-ledger profile yields 0.995 MWh/day of meter-capped payable response with zero meter-capped false payment, while no reserve job is paid before a later commitment. The trace-anchored N-1 number is a network-layer calibration identity; the independent end-to-end replay remains the stated performance boundary. Held-out calibration and AC panels delimit scope; deployment requires a signed ledger, co-located telemetry, topology access, and a capacity-safe workload-to-power conversion.

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